
HOMEWORK DAY 40 – *Average Value of a Function §5.5*

1. §5.5: 2

2. §5.5: 15

3. An aluminum rod is 8cm long. The temperature of the rod at a point x cm from one end is given by

$$T(x) = \begin{cases} x^3 - x^2 + 32, & 0 \leq x \leq 2 \\ 36 - 2x + x^2, & 2 < x \leq 8 \end{cases}$$

Find the average temperature of the rod.

4. §5.5: 20 (average vs maximal blood flow)

5. Consider the function $f(x) = \sin(x) - \cos(x)$.

(a) Find the average f_{av} of f over the interval $[a, a + \pi]$.

(b) For what value of $a \in [0, \pi]$ is the average on $[a, a + \pi]$ largest?

(c) The figure shows a graph of f . Show the interval $[a, a + \pi]$ over which the average is maximal and the line $y = f_{av}$ for that interval. Explain why for any other a the average would clearly be less.

